

1 Difference between Probability and Likelihood

Probability and likelihood are NOT the same. Probability is the chance of an outcome between 0 to 1. Likelihood is given dataset, which distributional parameter is most likely plausible. Likelihood is not between 0 and 1. Probability is MATHEMATICALLY equal to a likelihood, but not STATISTICALLY.

2 Frequentist Drawbacks

We Theoretically Estimate π using Maximum likelihood estimation (MLE) is the standard frequentist approach. We cannot incorporate our “expert” intuition, probabilistically speaking. How do we use side information, e.g., results of flips from other bottle caps? Asymptotic tools tend not to work well with small amounts of data. When distributional assumptions break down, the sampling distribution is messy, or the inferential target becomes more complex.

2.1 Definitions of Probability Equation

$P(L|T) = \frac{P(L \cap T)}{P(T)}$ $P(L|T)$: P of event L occurring given that event T has occurred. $P(L \cap T)$: P that both event L and event T occur. $P(T)$: P of the condition, event T , occurring on its own. new posterior $\propto$ new likelihood $\times$ old posterior

3 Bayesian Workflow

3.1 Likelihood and the "Product Rule"

Bayesians consider the population parameters as random. When we assume observations are (i.i.d.), the joint likelihood of our data is simply the product of individual probabilities.

If Independent: If you flip a coin 10 times, the likelihood of that sequence is $P(x_1|\theta) \times P(x_2|\theta) \cdots \times P(x_{10}|\theta)$. This makes the math easy to compute.

If Dependent: If the outcome of the second flip depends on the first (like drawing cards from a deck without replacement), you cannot just multiply them. You must use conditional probabilities: $P(x_1|\theta) \times P(x_2|x_1, \theta)$.

Latent variables are real but not directly observable quantities, which are linked to what we can measure. Instead, we observe partial signals and infer an underlying state.

Recursive Updating new evidence arrives, we update our beliefs, and yesterday's posterior becomes tomorrow's prior. We will update a probability using Bayes' rule, and we will start building the "prior"-posterior workflow.

3.2 Bayes Rule

The goal of Bayes' Rule is to find the Posterior Probability: $P(L|T) = \frac{P(T|L) \cdot P(L)}{P(T)}$ Prior: what we believe before new evidence (e.g., from a previous study).

Evidence term: how compatible the evidence is with each outcome (e.g., from an observed sample). Posterior: what we believe after incorporating the evidence. Hidden variables of interest are random (the prior distribution).

We use the conditional distribution of hidden variables given observations (the posterior distribution) to capture uncertainty.

F_M : "forgets ID on Monday," $P(F_M) = 0.10$.

F_T : "forgets ID on Tuesday," $P(F_T) = 0.10$.

F_M^c : "does not forget ID on Monday," $P(F_M^c) = 1 - 0.10 = 0.90$.

F_T^c : "does not forget ID on Tuesday," $P(F_T^c) = 1 - 0.10 = 0.90$.

We want $P(F_T^c | F_M^c)$. Start from the definition of conditional probability: $P(F_T^c | F_M^c) = \frac{P(F_T^c \cap F_M^c)}{P(F_M^c)}$. Use independence across days:

F_T^c and F_M^c are independent, so $P(F_T^c \cap F_M^c) = P(F_T^c) \times P(F_M^c) = 0.90 \times 0.90 = 0.81$.

Now, substitute into the conditional probability formula: $P(F_T^c | F_M^c) = \frac{0.81}{0.90} = 0.90$. So, the probability the student does not forget their ID on Tuesday, given they did not forget it on Monday, is 0.90. This is the same probability from Monday since both events are independent.

3.3 MAP and MLE

Maximum a posteriori (MAP) estimation is the Bayesian version of maximum likelihood estimation (MLE). MLE aims to find the value that maximizes the sample's likelihood function (i.e., we are just relying on our observed sampled data). MAP aims to obtain the value that maximizes our posterior distribution, which was obtained with our observed data (i.e., our evidence via the likelihood) weighted by our prior probability model.

4 Bayesian modelling involves these steps:

No p-values in bayesian models, p-value is a frequentist tools. As the data become more informative, the likelihood dominates, and the posterior becomes less sensitive to this prior choice. It is a big joint probability distribution.

Question: Pose a scientific question.

Design: Formulate variables and create a probabilistic model for them. Prior knowledge is included here!

Infer: Get posterior samples from the conditional distribution of any variable you want to infer, given your observed data and prior knowledge.

Check: Make sure the "samples" are actually from your posterior (these are model diagnostics, to be covered in week 4!)

Analyze: Use your samples to compute summaries: mean, maximum a posteriori (MAP), variance, posterior credible intervals (to be explained today), etc.

4.1 The Likelihood Selection (The "Design" Phase)

The first question is: What is the nature of your outcome variable?

- **Binary (Yes/No, Win/Loss)**
Likelihood: Bernoulli / Binomial Latent Variable: Probability of success (π) Bounds: $0 \leq \pi \leq 1$
- **Counts (0,1,2,...)** Likelihood: Poisson Latent Variable: Expected rate (λ) - continous Bounds: $\lambda > 0$
- **Continuous (No bounds)** Likelihood: Normal (Gaussian) Latent Variable: Mean (μ) Bounds: $-\infty < \mu < \infty$
- **Continuous (Positive only)** Likelihood: Log-Normal / Gamma Latent Variable: Shape (α) and Rate (β) Bounds: $\alpha > 0, \beta > 0$
- **Waiting Times** Likelihood: Exponential Latent Variable: Rate of decay (λ) Bounds: $\lambda > 0$

4.2 The Prior Selection (The "Question" Phase)

The prior represents your initial "latent" state (like Team Skill S_j).

Weakly Informative (The "Peer" Choice): Use Normal(0, 10) or Gamma(1, 1). These say, "I have a guess, but I'm open to being wrong."

Conjugate Priors (The "Math Shortcut"): Choose a prior that "matches" your likelihood so the posterior is in the same family.

Binomial Likelihood $\rightarrow$ Beta Prior

Poisson Likelihood $\rightarrow$ Gamma Prior

Normal Likelihood $\rightarrow$ Normal Prior

4.3 The Inference Method (The "Infer" Phase)

How will you actually calculate the posterior? When the prior is from the same family as the posterior, we call it a CONJUGATE PRIOR.

Grid Approximation: Use only for 1D problems. Conjugate

Math: Use when you need an instant, exact analytical answer and the model is simple. MCMC (Stan): Use for everything else. If you have multiple teams, hierarchical leagues, or complex predictors, you need a sampler like NUTS (No-U-Turn Sampler).

5 Beta-Binomial

For a posterior distribution where $\pi | Y = y \sim \text{Beta}(a_0 + y, b_0 + n - y)$, specifically Beta(16, 7):

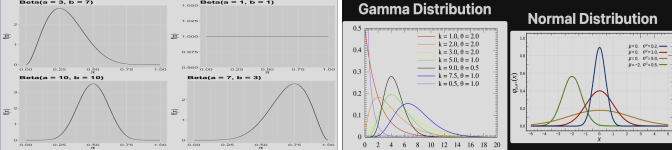
Posterior Mean: The calculation is $E(\pi | Y = y) = \frac{a_0 + y}{a_0 + b_0 + n}$

$n = 20$ free throws and scores $y = 14$

```
L <- qbeta(0.05, shape1 = 16, shape2 = 7)
data {
  int<lower=1> n;
  int<lower=0> X;
  #observed successes
  real<lower=0> a;
  real<lower=0> b;
  #hyperparams a and b
  parameters {
    real<lower=0, upper=1> pi;
    #Unknown param estimate
  }
  model {
    pi ~ beta(a, b);
    # 1. THE PRIOR
    X ~ binomial(n, pi);
    # 2. THE LIKELIHOOD
  }
  rocket_dictionary <- list(n = 20, X = 10, a = 1, b = 1)
  posterior_rockets <- sampling( # MCMC method of simulation
    object = rockets_stan, # 1. The compiled model
    data = rocket_dictionary, # 2. The input data list
    chains = 2, # 3. Number of independent scouts
    iter = 10000, # 4. Total steps per scout
    warmup = 1000, # 5. Learning the terrain (discarded)
    thin = 5, # 6. Saving every Nth step
    seed = 553, # 7. Reproducibility key
    algorithm = "NUTS", # 8. The navigation strategy
    round(qbeta(p = c(0.025, 0.975), shape1 = 42, shape2 = 18), 2)
```

Beta(a = 12, b = 8) is more informative because the distribution is more concentrated around 0.6, unlike prior Beta(a = 1, b = 1) that assigns the same density across all values of π . Advantage:

Because $n = 8$ is tiny, a prior lets the team encode realistic expectations about triage accuracy so the posterior for π does not swing wildly based on just a small handful of audited cases. This can lead to more stable staffing/safety decisions early in the rollout. **Disadvantage:** If the site really is different, a prior like Beta(a = 12, b = 8) can delay detection of poor performance by pulling the posterior upward toward “what usually happens as a prior knowledge,” which could cause the team to miss a safety problem in the first week and make overly optimistic decisions about adopting the checklist.



6 Gamma-Poisson

Posterior estimate can be a location summary (i.e., planning around a rate) such as the posterior mean $\lambda | Y = y \sim \text{Gamma}(s', r')$, $E(\lambda | Y = y) = \frac{s'}{r'}$

Posterior estimate can be a spread summary (i.e., an uncertainty measure) such as the posterior standard deviation $\lambda | Y = y \sim \text{Gamma}(s', r')$, $sd(\lambda | Y = y) = \frac{\sqrt{s'}}{r'}$

Gamma(s', r'), $sd(\lambda | Y = y) = \frac{\sqrt{s'}}{r'}$
Posterior summary can be an extreme-value summary (high but plausible rate) such as a high posterior quantile. For instance, define the 0.95 posterior quantile.

```
data {
  int<lower=0> n;
  int<lower=0> t;
  array[n] int<lower=1> win_team;
  array[n] int<lower=1> lose_team;
  array[n] int<lower=0> score_diff;
}
# score differences
parameters {
  vector<lower=0>[t] skill;
}
# real-valued vector
model {
  skill ~ gamma(1, 1);
  #Vectorized prior: Gamma(1,1)
  for (i in 1:n) {
    int win_index = win_team[i];
    int lose_index = lose_team[i];
    score_diff[i] ~ poisson(log(1 + exp(skill[win_index] - skill[lose_index])));
  }
}
```

7 Monte Carlo Algorithm

You have a closed analytical form for the posterior distribution $f(\theta | y)$. You can build your independent Monte Carlo sample $\{\theta_1, \dots, \theta_N\}$ of size N , from $f(\theta | y)$ by selecting each element θ_i ($i = 1, \dots, N$) in the following way: **Step 1.** Draw θ_i from $f(\theta | y)$. **Step 2.** Go there. This is a random walk!

8 Markov Chain

In terms of our Bayesian modelling for the posterior, it is a sequence $\{\theta_1, \dots, \theta_N\}$ of size N whose elements are **NOT ENTIRELY INDEPENDENT** (hence the word chain!). It implicates the so-called **Markov property**: The next sampled θ_{i+1} will depend on the current θ_i but not on previous $\theta_{i-1}, \theta_{i-2}, \dots$. More formally, because of this property, the conditional probability function of θ_{i+1} can be depicted as: $f(\theta_{i+1} | \theta_i, y) = f(\theta_{i+1} | \theta_1, \theta_2, \dots, \theta_i, y)$.

9 MCMC

The approximate posterior's width is determined by the model (our likelihood and any noise/variability assumptions), the prior (how concentrated your beliefs are before seeing data), and the observed data (how much information the data contain, often more observations or less noisy measurements make the likelihood more concentrated). In contrast, MCMC settings like warmup length, iterations, and number of chains do not change the target posterior; they mainly affect whether the chains explore it adequately and whether the resulting posterior summaries and plots are stable and trustworthy

9.1 Disadvantage of MCMC

MCMC too slow or too resource-intensive for a 10-minute operational decision, whereas closed-form posteriors give instant calculation. With MCMC in a 20-parameter model, chains can mix slowly, parameters can be correlated, and exploration can be uneven, so you need diagnostics to confirm the approximate posterior samples are reliable. Under time pressure, skipping those checks can lead to misleading summaries and decisions. With many 20 covariates, small modeling choices can noticeably change the posterior. MCMC will faithfully reflect whatever you specified, but under time pressure you may not have time to stress-test those choices, whereas closed-form setups are often used in simpler models with fewer moving parts.